propriate deep-sea lines, it is possible to remove the tag of critically polluted areas from these clusters," noted a Gujarat Chamber of Commerce and Industry (GCCI) report while making a case for marine outfalls. This will "allow for [industrial] expansion in these regions and create direct and indirect employment opportunities...generating revenue for the government," it added. The proposal, though approved by the industries department, was rejected by the water resources department. The reason being that the outfalls were proposed at the site where the Gujarat government plans to undertake its ambitious Kalpasar project—a freshwater reservoir envisaged for tidal power generation, drinking and irrigation purposes—by building a 30-km-long dam across the Gulf of Khambhat. (First proposed in 1988, the project is still undergoing feasibility studies.)

Gujarat Water Infrastructure Limited and Gujarat Pollution Control Board (GPCB) are the executing agencies.

The total cost of construction is pegged at Rs 8,321 crore, of which the participating industries will bear 20-30 per cent. The pipelines are expected to become operational over the next 10 years, though their offshore length and depth are still being worked out. In the case of the Ahmedabad pipeline, which will cost about Rs 1,500 crore, 3,050 units from 11 industrial areas have joined the project. "Meeting the norms set for effluent discharge into rivers is financially unviable for the industry," says former GCCI chairman Shailesh Patwari, while emphasising the project's importance. "These norms are very strict globally as river water percolates into aquifers, and is used for irrigation and drinking. Moreover, not being perennial, the Sabarmati and other rivers fail to dilute the industrial discharge." So, the industries in Gujarat, he adds, are left with two options: deep-sea discharge or zero liquid discharge. The latter process, wherein all water is recovered and contaminants reduced to solids, "is not financially viable for small- and medium-scale industries", says Patwari. "So, deep-sea discharge is the only solution."

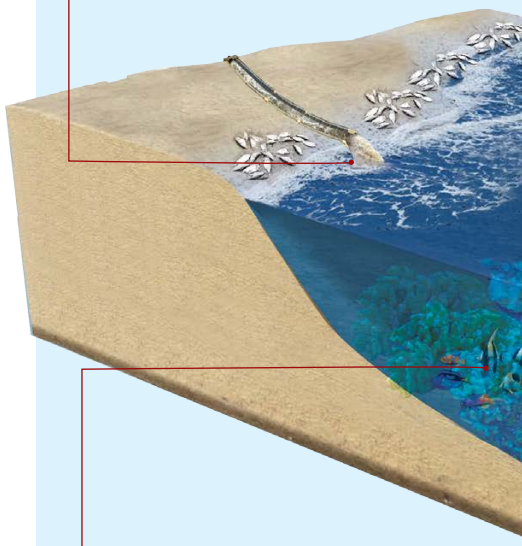


"NORMS FOR DICHARGING EFFLUENTS INTO THE RIVERS ARE VERY STRICT. MEETING THEM IS UNVIABLE. DEEP-SEA DISCHARGE IS THE ONLY SOLUTION"

SHAILESH PATWARI, *Former Chairman, Gujarat Chamber of Commerce and Industry*

ECOLOGICAL PRICE

A study published in the journal *Frontiers in Environmental Science* in June 2022 highlighted the decline in fish stock due to pollution caused by marine outfalls in Doha Bay along Qatar's capital



A study published in Aug. 2020 found a 45 per cent decline in biomass of important fish species in West Hawaii's coral reefs over a 10-year period, owing mainly to sewage pollution

THE GRAND PLAN

So, while the marine outfall proposal hit a roadblock, the Gujarat High Court had been repeatedly taking the state government to task over the past few years for its inability to clean the Sabarmati. In June this year, the government was directed to submit a concrete time-bound action plan in this regard. A month later, the water resources department finally gave its nod to the deep-sea effluent discharge project, but only after the total length of the pipeline originating in Ahmedabad was extended from 130 km to 210 km to take its outfall point beyond the proposed Kalpasar lake. Following this, the other six pipelines have also received an in-principle approval. The

A CAUSE OF CONCERN

The world's first marine outfall, built in Santa Monica, United States, dates back to 1910. The largest, in Boston, discharges sewage after extensive treatment. Over 100 marine outfalls dot India's 7,516-km coastline as well, but several reports have raised concern over the bulk of wastewater being disposed of without being treated properly, putting marine ecosystems at potential risk.

Despite the sea's assimilation abilities, chemical-infested effluents are known to reduce the oxygen content in water, impacting marine life. Studies conducted by the Chennai-based National Centre for Coastal Research (NCCR), which works with state pollution control boards to monitor the effluent discharge at about 50 locations in India, have shown that the fish stock has taken a beating due to marine pollution, impacting the fishing industry over the course of time. "The Indian coastal waters do